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1. I. a. $E[X] = 1 \cdot 0.6 + 0 \cdot 0.4 = 0.6$

$$\text{Var}(X) = pq = 0.4 \cdot 0.6 = 0.24$$

$$E[Y] = E[3X+5] = 3E[X] + 5 = 3 \cdot 0.6 + 5 = 1.8 + 5 = \boxed{6.8}$$

$$\text{Var}(Y) = \text{Var}(3X+5) = 9\text{Var}(X) = 9 \cdot 0.24 = \boxed{2.16}$$

b. $X \sim N(0, 1)$ $E[X] = 0$ $\text{Var}(X) = 1$

$$E[Y] = E[2X+3] = 2E[X] + 3 = \boxed{3}$$

$$\text{Var}(Y) = \text{Var}(2X+3) = 4\text{Var}(X) = \boxed{4}$$

II. a. $P(\text{spam}) = 0.2$ $P(\text{offer} | \text{spam}) = 0.4$ $P(\text{offer} | \text{spam}^c) = 0.05$

$$P(\text{spam} | \text{offer}) = \frac{P(\text{offer} | \text{spam}) \cdot P(\text{spam})}{P(\text{offer})}$$

$$\begin{aligned} P(\text{offer}) &= P(\text{offer} | \text{spam}) \cdot P(\text{spam}) + P(\text{offer} | \text{spam}^c) \cdot P(\text{spam}^c) \\ &= 0.4 \cdot 0.2 + 0.05 \cdot 0.8 \\ &= 0.08 + 0.04 = 0.12 \end{aligned}$$

$$P(\text{spam} | \text{offer}) = \frac{0.4 \cdot 0.2}{0.12} = \boxed{\frac{2}{3}}$$

b. $P(A) = \frac{1}{2}$ $P(B) = \frac{1}{2}$ $P(r | A) = 0.4$ $P(b | A) = 0.6$

$$P(r | B) = 0.8$$
 $P(b | B) = 0.2$

$$P(B | r) = \frac{P(r | B) \cdot P(B)}{P(r)}$$

$$P(r) = P(r | B) \cdot P(B) + P(r | A) \cdot P(A)$$

$$= \frac{1}{2} (0.8 + 0.4) = \frac{1}{2} (1.2) = 0.6$$

$$P(B | r) = \frac{0.8 \cdot \frac{1}{2}}{0.6} = \frac{0.8}{1.2} = \boxed{\frac{2}{3}}$$

III. a. Entropy measures how uncertain or random a distribution is

- high entropy for equally likely outcomes

$$-H(P) = -\sum_i p_i \log p_i$$

Cross entropy measures how well distribution Q approximates the true distribution P

- if $Q = P$, cross entropy equals regular entropy

- the worse Q approximates P , the higher the cross entropy

$$H(P, Q) = - \sum_i p_i \log q_i$$

KL-Divergence measures the "distance" between two distributions

- how much information is lost when you use Q to approximate P

$$D_{KL}(P \parallel Q) = \sum_i p_i \log \frac{p_i}{q_i}$$

$$b. H(P, Q) = H(P) + D_{KL}(P \parallel Q)$$

$$2. I. a. \|A\|_F = \sqrt{1^2 + 2^2 + 3^2 + 4^2} = \sqrt{1+4+9+16} = \sqrt{30} = \sqrt{30}$$

$$b. \|B\|_1 = m \times (n, 6) = 6$$

$$c. \|C\|_\infty = 5$$

$$II. a. AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 2+2 & 0+6 \\ 6+4 & 0+12 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 10 & 12 \end{bmatrix}$$

b. No, Matrix multiplication is not commutative

$$BA = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2+0 & 4+0 \\ 1+9 & 2+12 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 10 & 14 \end{bmatrix}$$

$$AB \neq BA$$

$$c. Ax = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ -1 \end{bmatrix} = \begin{bmatrix} 8-1 \\ 0-3 \end{bmatrix} = \begin{bmatrix} 7 \\ -3 \end{bmatrix}$$

$$III. a. \frac{\partial f}{\partial x} = 6x + 2y \quad \frac{\partial f}{\partial y} = 2x + 2y$$

$$\nabla f = \langle 6x + 2y, 2x + 2y \rangle$$

$$b. \|Ax - b\|_2^2 = (Ax - b)^T (Ax - b) = (x^T A^T - b^T) (Ax - b)$$

$$= x^T A^T A x - x^T A^T b - b^T A x + b^T b$$

$$x^T A^T b = (x^T A^T b)^T \text{ because } x^T A^T b \text{ is a scalar}$$

$$(x^T A^T b)^T = b^T A x$$

$$\Rightarrow \|Ax - b\|_2^2 = x^T A^T A x - 2b^T A x + b^T b$$

$$\nabla_x (x^T C x) = 2Cx \text{ when } C \text{ is symmetric } (A^T A \text{ is symmetric})$$

$$\nabla_x (b^T A x) = A^T b$$

$$\nabla_x (b^T b) = 0$$

$$\nabla_x f(x) = 2A^T A x - 2A^T b = \boxed{2A^T(Ax - b)}$$

$$c. w_x = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - x_2 \\ 2x_2 \end{bmatrix}$$

$$w_x([1]) = \begin{bmatrix} 0 \\ 2 \end{bmatrix} \quad \sigma(w_x) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$f(x) = v^T \sigma(w_x)$$

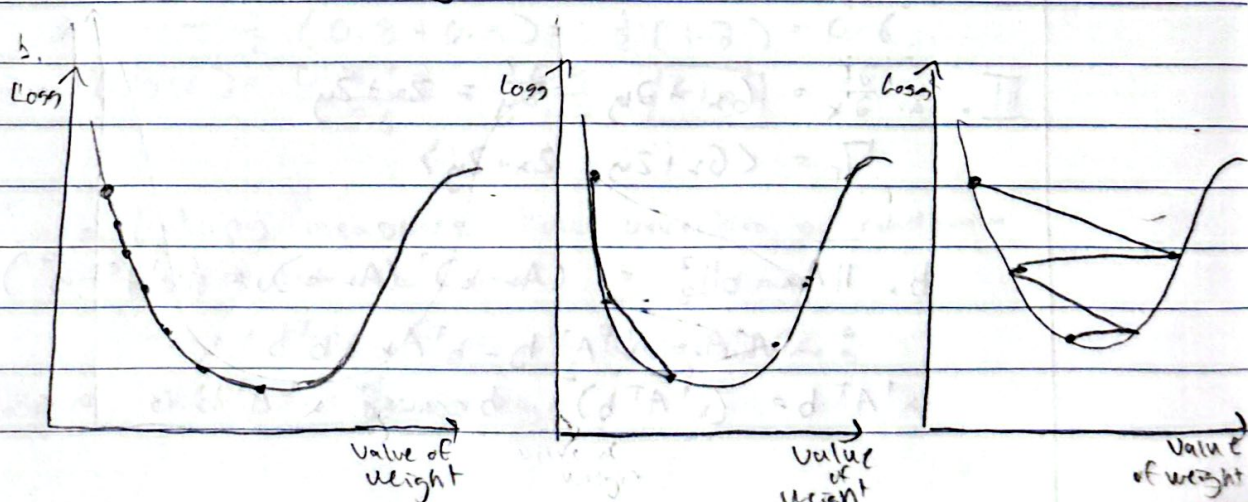
$$(f'(x)) = w^T \cdot \text{diag}(\sigma'(w_x)) \cdot v$$

$$\sigma'(w_x) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{diag}(\sigma'(w_x)) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$f'([1]) = \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \boxed{\begin{bmatrix} 0 \\ -4 \end{bmatrix}}$$

3. I. a. No, since the term is grad^2 , this update rule always decreases w regardless of the sign of the gradient, so if initial weight is to the left of the minimum, it will move further away from it.



II. a. Model A is likely overfitting because of the large difference in training and validation accuracy, while Model B doesn't have that big of a difference between these two metrics making it better for generalization.

b. The specific phenomenon is called overfitting, while training when training loss & validation loss start off moving in the same direction and then diverge.

1. Early stopping - stop training at the point where validation loss is at its minimum, before it starts rising again.

2. Reduce model complexity - use a simpler model with fewer parameters so it can't memorize the training data as easily.

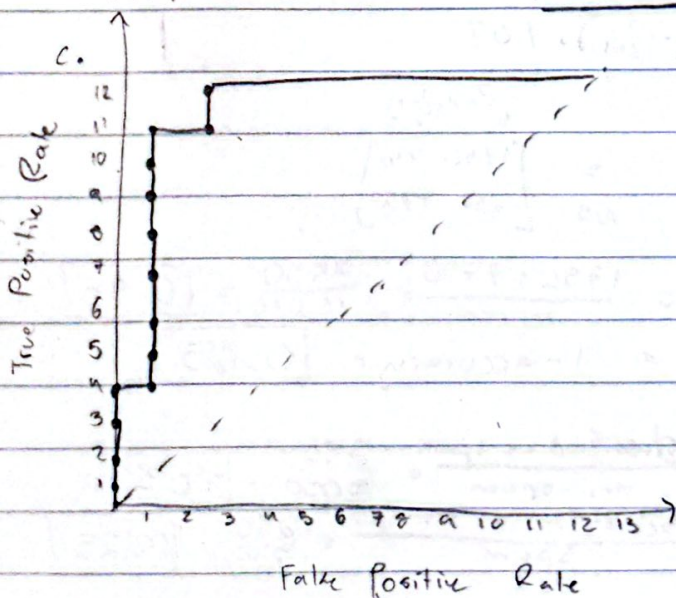
4. I. a. There are 9 ranking errors

- x_{13} appears before 8 positive samples

- x_{15} appears before 1 positive sample

b. $12 \times 13 = 156$ total pairs

$$\text{error rate} = 9/156 \approx 0.058$$



$$I. a. L_{0-1} = \begin{cases} 1 & m < 0 \\ 0 & m > 0 \end{cases} \begin{matrix} \text{(misclassification)} \\ \text{(correctly classified)} \end{matrix}$$

$$L_{\text{hinge}} = \max(0, 1-m) = \begin{cases} 0 & m \geq 1 \text{ (confident, correct)} \\ 1-m & m < 1 \text{ (unconfident, incorrect)} \end{cases}$$

$$L_{\text{squared}} = \begin{cases} 0 & m \geq 1 \\ (1-m)^2 & m < 1 \end{cases}$$

$$b. 1. m = \frac{3 \cdot 2 + 2 \cdot 2 + 4 \cdot 3 - 18}{9 + 4 + 16} \cdot 1 = \frac{6 + 4 + 12 - 18}{13 + 16} = \frac{10 - 6}{29} = \frac{4}{29}$$

$$L_{0-1}(4/29) = 0$$

$$L_{\text{hinge}}(4/29) = 25/29$$

$$L_{\text{squared}}(4/29) = 0.743$$

$$2. m = \frac{3 \cdot 3 + 2 \cdot 3 + 4 \cdot 1 - 18}{9 + 4 + 16} \cdot -1 = \frac{9 + 6 + 4 - 18}{29} \cdot -1 = \frac{15 - 14}{29} \cdot -1 = -\frac{1}{29}$$

$$L_{0-1}(-\frac{1}{29}) = 1$$

$$L_{\text{hinge}}(-\frac{1}{29}) = 30/29$$

$$L_{\text{squared}}(-\frac{1}{29}) = 1.07$$

$$III. a. C = \begin{matrix} & \text{Predicted} \\ & \begin{matrix} \text{Spam} & \text{Not Spam} \end{matrix} \\ \text{Actual} & \begin{bmatrix} 1750 & 250 \\ 250 & 7750 \end{bmatrix} \end{matrix}$$

$$\text{Accuracy} = \frac{1750 + 7750}{10,000} = \frac{9500}{10,000} = \boxed{0.95}$$

$$\text{error rate} = 1 - \text{accuracy} = \boxed{0.05}$$

$$b. \text{FPR} = \frac{\text{misclassified as spam}}{\text{non-spam}} = \frac{250}{8000} = \boxed{0.031}$$

$$\text{FNR} = \frac{\text{misclassified as non-spam}}{\text{spam}} = \frac{250}{2000} = \boxed{0.125}$$

$$c. \text{ TPR} = \frac{\text{correctly classified spam}}{\text{spam}} = \frac{1750}{2000} = 0.875$$

$$\text{TNR} = \frac{\text{correctly classified non-spam}}{\text{non-spam}} = \frac{7750}{8000} = 0.969$$

$$d. \text{ Precision} = \frac{1750}{2000} = 0.875 \quad (\text{correctly predicted spam out of all classified as spam})$$

$$F_1 = 2 \cdot \frac{\text{Recall} \cdot \text{Precision}}{\text{Recall} + \text{Precision}} = 2 \cdot \frac{0.875 \cdot 0.875}{0.875 + 0.875} = 0.875$$

